



CSE-476: Introduction to Natural Language Processing
Quiz 5

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. Which option best describes instruction tuning?
- (A) Pre-training on raw text with a next-word loss
 - (B) Post-training on instruction–response pairs to follow tasks
 - (C) Adding more layers to reduce perplexity values
 - (D) Using masks to block attention to future tokens
- Q2. Which benchmark is a widely used multiple-choice QA test for LLMs?
- (A) HotpotQA (generative multi-hop answers)
 - (B) HumanEval (code generation via execution)
 - (C) SQuAD (extractive span selection)
 - (D) MMLU (multiple-choice benchmark)
- Q3. A model answers 7 to “A farmer has 10 sheep; all but 3 run away.” What type of hallucination is this?
- (A) Factual hallucination (wrong world facts)
 - (B) Commonsense hallucination (unsafe commonsense)
 - (C) Reasoning hallucination (faulty logical steps)
 - (D) No hallucination: answer is correct
- Q4. Why use pairwise human preferences for alignment after instruction tuning?
- (A) They’re easier than absolute “goodness” scores and capture relative quality
 - (B) They strictly reduce compute and memory costs
 - (C) They guarantee perfect annotator fairness
 - (D) They remove the need for any supervised data
- Q5. In PPO-based RLHF, what is the reward model’s job?
- (A) Set attention masks during decoding
 - (B) Collect human preference pairs online

- (C) Tune optimizer hyper-parameters each step
 - (D) Score whole model responses so PPO can assign rewards
- Q6. What is a key difference between DPO and PPO-based RLHF?
- (A) DPO trains directly on preference pairs without a separate reward model
 - (B) DPO requires a value network to compute advantages
 - (C) DPO prohibits using preference data
 - (D) DPO only optimizes next-word likelihood
- Q7. What primary issue is RAG introduced to help with?
- (A) Grounding generations by retrieving external sources
 - (B) Increasing the parameter count to store more facts
 - (C) Replacing tokenization with sub-word merges
 - (D) Eliminating the need for instruction tuning
- Q8. In the retrieval primer, what does BM25 add beyond plain TF-IDF?
- (A) Backpropagates through retrieval to update encoders
 - (B) Byte-pair encoding to compress queries
 - (C) Learned neural embeddings for queries and documents
 - (D) Document-length normalization and tunable term saturation on term frequency
- Q9. What does Recall@k measure?
- (A) All retrieved docs
 - (B) Relevant docs retrieved out of all relevant ones
 - (C) Model accuracy
 - (D) Token overlap
- Q10. In Dense Passage Retrieval (DPR), what are the Query Encoder and Passage Encoder typically based on?
- (A) RNNs
 - (B) Decision Trees
 - (C) BERT-like models
 - (D) Linear Regression

End of Paper